INCH-POUND

MIL-C-20354C (AR)
AMENDMENT 3
2 October 2000
SUPERSEDING
AMENDMENT 2
23 June 1994

MILITARY SPECIFICATION

CARTRIDGE, 105MM, ILLUMINATING, M314A3, LOADING, ASSEMBLING AND PACKAGING

MIL-C-20354C was inactivated after 12 August 1997 for new design.

This amendment forms a part of MIL-C-20354C (AR), dated 19 May 1988, and is approved for use by the U.S. Army Tank-automotive and Armaments Command, Armament Research, Development and Engineering Center and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 2.1.1 Delete the following under Specifications Military:

"MIL-A-2550 - Ammunition, and Special Weapons General Specification for,

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for,"

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* 2.1.1 Delete the following under Standards, Military:

"MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 - Quality Assurance Terms and Definitions

MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification

MIL-STD-1167 - Ammunition Data Card

MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes"

* 2.1.1 Add the following under Standard MILITARY:

"MIL-STD-1168- Lot Numbering of Ammunition and Ammunition Data Card

MIL-STD-1916 - DOD Preferred Methods for Acceptance of Product"

2.1.2, Other Government documents, drawings, and publications. Delete drawing 7549072 in its entirety and replace with:

"12934901 - Packing and marking for container ammo metal, PA117, for 105mm Howitzer ctg. Packed in PA55 or M105A3

19-48-4231/- 42-20PM1006 Unitization procedures for ammunition components packed in a cylindrical metal or plastic container on a 4-way entry metal pallet"

Change "M105A2" to "M105A3" wherever it appears.

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Delete under inspection equipment drawings:

"9201136 - Tunnel, Light
9201390 - Procedure, Photocell Checkout
9200964 - Photocell
9200964 - Photometer Head with Cell"

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* 2.2 Other Publications: Change address for ASTM from: "1916 Race Street, Philadelphia, PA 19103"

TO: "100 Barr Harbor Drive, Conshohocken, PA"

* 2.2 Other Publications. Add the following new

Publication before the paragraph heading "CODE OF FEDERAL REGULATIONS."

"AEROSPACE INDUSTIRES ASSOCIATION OF AMERICA, Inc. (AIA/NAS)

NAS-410 - NAS Certification & Qualification of Nondestructive Test Personnel

(Application for copies should be addressed to the Aerospace Industries Association of America, Inc. (AIA/NAS), 1250 Eye Street, NW; Suit 1100 Washington, DC 20005.)"

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3.2.3, Burning time – candle power of the illuminant assembly (Dwg. 9206749).: Delete in its entirety and replace with:

"3.2.3 Luminosity and burning time. The average luminosity of the canister assemblies shall be not less than 525,000 candlepower. The burning time of the canister assemblies shall be not less than 60 seconds. Determination shall be as specified in 4.4.3.5 and 4.5.3."

* 3.5.2 Delete "to a .40% AQL," in last sentence and substitute "to a level IV."

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* 3.6.1 Delete the following part of sentence:

"and shall be in accordance with the provisions of MIL-A-2550"

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* 4.3.2 Delete "MIL-A-48078" and change to "MIL-STD-1916"

* 4.3.3 Change "MIL-A-48078" to "MIL-STD-1916."

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TABLE 1, First Article Inspection: Add the following new inspections to Illuminating Canister:

"Illuminant composition test	4 ounces	3.2.1	4.5.2.2/ 4.5.2.3
Luminosity and burning Time	5 pair	3.2.3	4.4.3.5/ 4.5.3"

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TABLE 1, First Article Inspection: First Fire Composition: Delete in its entirety and replace with:

"First Fire Composition First Fire Composition Test	4 ounces	3.2.1	4.5.2.1/ 4.5.2.3"

Add the following new first article tests:

"Container, Ammo. Fiber, M105A3, for 105mm Howitzer Ammunition (Dwg. 7549073) Examination for defects	5	3.1	4.2.2.11/ 4.2.2.12

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Packing and Marking for Container Ammo Metal, PA117, for 105mm Howitzer Ctg. Packed in PA55 or M105A3 (Dwg. 12934901)			
Examination for defects	5	3.1	4.2.2.13
Palletized Projectile, M314A3; Packed in accordance with drawing 19-48-4231/ 45-20PM1006			
Examination for defects	1 pallet	-	4.2.2.14”

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* 4.4.1 Delete “MIL-A-48078” and substitute “MIL-STD-1916.”

* 4.4.2 Delete “MIL-A-48078” and substitute “MIL-STD-1916.”

a. Delete: “with MIL-STD-1235 or MIL-STD-105 as applicable,” and substitute:

“with MIL-STD-1916,”

PAGES 10 THRU 20 and 23

* 4.4.2.1 Thru 4.4.2.10 and 4.4.2.13, Quality Conformance Inspection TABLES. Delete “0.40%” and “0.65%” AQLs, and substitute “Level IV” for major defects, and “Level II” for minor defects.

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* 4.2.2.7 Add the following Major Defect:

Major			
“108. Plug disassembly torque, max. (if applicable)	Level IV	3.1	gage”

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* 4.2.2.11 Delete in its entirety and replace with new paragraph 4.2.2.11 (included).

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* 4.2.2.12 Delete in its entirety and replace with new paragraph 4.2.2.12 (included).

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* Add new paragraphs 4.2.2.14 and 4.2.2.15 (included).

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CLASSIFICATION OF CHARACTERISTICS

4.2.2.11	TITLE Unsealed Fiber Container	SHEET 1 OF 1		DRAWING NUMBER 754073
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 754073 (sealed) INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Packing component missing, damaged, or improperly assembled	Level IV	3.1	Visual
102	Cartridge cannot be inserted or removed by hand	Level IV	3.1	Manual
103	Container covers cannot be removed by hand	Level IV	3.1	Manual
104	Insufficient fillers for tight pack	Level IV	3.1	Gage
Minor				
201	Evidence of poor workmanship	Level II	3.6	Visual/Manual

NOTES:

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4.2.2.12	TITLE: Sealed fiber container	SHEET 1 OF 1		DRAWING NUMBER: 7549073
				NEXT HIGHER ASSEMBLY: 12934901
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Sealing strip incomplete	Level IV	3.1	Visual
102	Sealing strip wrinkled	Level IV	3.1	Visual
103	Sealing strip material improper	Level IV	3.1	Visual/Certification
104	Sealing strip damaged	Level IV	3.1	Visual
105	Interior fiber moisture content excessive at time of pack	Level IV	3.1	Gage
106	Component missing	Level IV	3.1	Visual
107	Marking misleading, missing, incorrect, or illegible	Level IV	3.1	Visual
108	Tight pack	Level IV	3.1	Manual/ Audible
Minor				
201	Tear tab length below minimum	Level II	3.1	Gage/SME
202	Component damaged	Level II	3.1	Visual
203	Evidence of poor workmanship	Level II	3.6	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

4.2.2.14	TITLE: Sealed metal container	SHEET 1 OF 1		DRAWING NUMBER: 12934901
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY / INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Container damaged exposing contents	Level IV	3.1	Visual
102	Metallic seal missing or improperly secured	Level IV	3.1	Visual/ Manual
103	DoD symbol missing or incorrect	Level IV	3.1	Visual
104	Air leak test	100%	3.1	4.5.4
105	Screw not properly replaced and torqued after test	Level IV	3.1	Visual/ Gage
106	Contents loose inside metal container	Level IV	3.1	Manual/ Audible
107	Interior packing material moisture content excessive at time of pack	Level IV	3.1	Gage
108	Cloth strap incorrect width or length	Level IV	3.1	SME/Gage
109	Cloth strap assembled incorrectly	Level IV	3.1	Visual
110	Cloth strap missing	Level IV	3.1	Visual
111	Marking incorrect, missing, or unidentifiable	Level IV	3.1	Visual
Minor				
201	Evidence of poor workmanship	Level II	3.6	Manual/ Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

	TITLE			DRAWING NUMBER
4.2.2.15	Palletized Projectile, M314A3; packed in accordance with drawing 19-48-4231/45-20PM1006	SHEET 1 OF 1		
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Strapping missing	Note 1	3.6	Visual
102	Strapping loose	Note 1	3.6	Visual/Manual
103	Incorrect number of containers on pallet (except for light load)	Note 1	3.6	Visual
104	Containers not all positioned in the same direction	Level IV Note 2	3.6	Visual
105	Containers not properly interlocked	Level IV Note 2	3.6	Visual
106	Marking missing, incorrect, or illegible	Level IV Note 2	3.6	Visual
Minor				
101	Evidence of poor workmanship	Level II Note 2	3.6	Visual/ Manual

NOTES:
Note 1 – Sample size to be the smallest number of pallets that contain a quantity of projectiles that reflects Level IV.
Note 2 – If inspection requires pallets to be opened, the entire quantity of projectiles on that pallet may, at the Government inspector’s option, be applied to the sample quantity, provided that the pallet is selected randomly.

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4.4.3.5 Static test of illuminant canister assembly. Delete in Its entirety and replace with:

"4.4.3.5 Luminosity and burning time (see 3.2.3 and drawing 9206749). The test shall be performed as specified in 4.5.3, and in accordance with Dwg. 9206749. A sample of two canister assemblies shall be selected in accordance with the sampling plan shown in 4.4.3.5.1. One canister assembly shall be tested for static luminosity and burning time. If the luminosity is less than 525,000 candlepower, the second candle shall also be tested. If the resulting average of the combined sample is less than 525,000 candlepower (luminosity), inspection lots represented by the samples shall be rejected (see 4.4.3.5.1.7). No one sample can exhibit a burning time of less than 60 seconds. The candlepower integration time shall be from the best continuous 55 seconds of burn time."

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4.4.3.5.1: Delete in its entirety and replace wilth:

"4.4.3.5.1 Sampling plan for luminosity and burning time. An inspection lot for the canister assembly shall be eighty-five candles produced consecutively from one mixing and pressing operation, and will be designated as one mix/batch for identification purposes (see 6.8). Smaller lot sizes will be accepted when dictated by interruption or shutdown of operations. The illuminant, as it is mixed, shall be dispensed into containers which shall be numbered and used in order."

4.4.3.5.1.1: Delete in its entirety and replace with:

"4.4.3.5.1.1 Level one sampling. A sample of two candles from each inspection lot shall be selected at random. The canister assembly shall be tested for compliance with the requirements specified in paragraph 3.2.3. This sampling frequency shall be continued until fifteen consecutive samples of two candles comply with the requirements of paragraph 3.2.3."

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4.4.3.5.1.2 Delete in its entirety and replace with:

"4.4.3.5.1.2 Level two sampling. After successful completion of level one sampling, the sampling frequency may be reduced to two samples from each group of six inspection lots. The canister assemblies shall be tested for compliance with the requirements specified in paragraph 3.2.3. This sampling frequency shall be continued until fifteen consecutive samples of two candles comply with the requirements of paragraph 3.2.3. In the event that one lot fails to comply with the requirements, the sampling shall revert to level one sampling."

4.4.3.5.1.3: Delete in its entirety and replace with:

"4.5.3.5.1.3 Level three sampling. After successful completion of level two sampling, the sampling frequency may be reduced to two samples from each group of fifteen inspection lots. The canister assemblies shall be tested for compliance with the requirements specified in paragraph 3.2.3. This sampling frequency shall be continued until fifteen consecutive samples of two candles comply with the requirements of paragraph 3.2.3. In the event that one lot fails to comply with the requirements, the sampling shall revert to level two sampling."

4.4.3.5.1.4: Delete in its entirety and replace with:

"4.4.3.5.1.4 Level four sampling. After successful completion of level three sampling, the sampling frequency may be reduced to two samples from each shift (not to exceed ten hours) of production. The canister assemblies shall be tested for compliance with the requirements specified in paragraph 3.2.3. In the event that one lot fails to comply with the requirements, the sampling shall revert to level three sampling."

4.4.3.5.1.5: Delete in its entirety and replace with:

"4.4.3.5.1.5 In the event of any production stoppage greater than two weeks, the sampling plan shall revert to level one, with only seven consecutive samples required to advance to each reduced level until the level of sampling in use when production was stopped is reached."

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4.4.3.5.1.6: Delete in its entirety and replace with:

"4.4.3.5.1.6. If the mixing operation has not been shut down, but merely diverted to another item, and has not experienced any cause for rejection per the sampling plan of that item(s), the sampling level shall remain unchanged, and fifteen consecutive successful samples of two candles shall be required to advance to the next level, if applicable."

4.4.3.5.1. Sampling plan for luminosity and burning time: Add new subparagraph 4.4.3.5.1.7:

"4.4.3.5.1.7 In the event of failure of any sample to comply with the requirements, each individual inspection lot within the mix/batch must be tested. Failure of an inspection lot to comply with the requirements shall result in rejection of that particular inspection lot."

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* 4.4.4 Delete last sentence in its entirety and substitute the following:

"See paragraph 6.3 herein."

* 4.4.4.2 Delete entire paragraph and substitute the following:

"4.4.4.2 Certification on Non-Destructive Test (NDT) Personnel.

All personnel operating or calibrating NDT equipment shall be qualified and certified in accordance with NAS-410."

4.5.2.1 First Fire Composition: Change "Equal parts by volume of carbon tetrachloride and anhydrous methanol" to "Equal parts by volume of dimethylformamide and anhydrous methanol".

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* 4.5.3. Delete in its entirety and replace with:

"4.5.3 Luminosity and burning time. The discs shall be removed from the candle prior to testing. In calculating the candle power of any one illuminant assembly, the best continuous 55 seconds of illuminometer readings are to be used. The test shall be conducted utilizing equipment and procedures as specified on drawing 9201268 or Government approved equivalent."

Add new paragraph 4.5.4 to follow paragraph 4.5.3:

"4.5.4 Air pressure test. The packed metal container assemblies shall be subjected to the internal air pressure test as specified on drawing 12934901. Observation shall be made for a pressure drop exceeding the requirement for the specified time. The cartridge packed in a defective container may be removed and packed in a replacement container to be tested."

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5.1.1 Change the first sentence to read "Each cartridge shall be packaged in accordance with drawing 7549073."

5.2.1 Delete the first sentence and replace with "Each packaged Fiber container (Dwg. 7549073) shall be packed in a PA117 metal container and cushion assembly in accordance with drawing 12934901.".

Add new paragraph 5.2.2:

"5.2.2. Level A – Alternative. Only with the specific approval of the contracting officer, the cartridge in fiber container shall be packed in a wooden box in accordance with dwg. 7549072."

5.3 Delete in its entirety and replace with the following:

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"5.3 Marking.

5.3.1 Level A. The fiber container shall be marked in accordance with drawing 7549073.

5.3.2 Level A. The metal container outer pack shall be marked in accordance with drawings 12934901 and 12982865.

5.3.3 Level A (wood box). The wood box shall be marked in accordance with drawings 7549073 and 12982865.

Add new paragraph 5.4 to follow 5.3:

"5.4 Unitization.

5.4.1 Level A. The packed metal container shall be unitized in accordance with drawing 19-48-4231/45-20PM1006."

* 6.2 Delete entire paragraph and substitute the following:

"6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification

b. Packaging requirements (see 5).

c. Requirements for submission of first article sample (see 4.3).

d. Government Furnished Equipment.

d. Requirements for submittal of Acceptance Inspection Equipment.

f. Requirements for ammunition lot numbering

g. Requirements for Certification of Nondestructive Test (NDT) personnel (see 6.4)."

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* 6.3 Delete the entire paragraph and substitute the following:

"6.3 Submission of inspection procedures and equipment designs for approval. Submit equipment designs, as required, to Commander, TACOM-ARDEC, ATTN: AMSTA-AR-QAA-R, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contractor Data Requirements List, DD Form 1423 in the contract."

* 6.4 Delete paragraph 6.4 in its entirety and substitute:

"6.4 Certification of nondestructive test (NDT) personnel. All personnel operating or calibrating NDT equipment shall be qualified and certified in accordance with NAS-410. Qualification to other standards must be submitted to and approved by Commander, ARDEC, ATTN: AMSTA-AR-QAA-R, Picatinny Arsenal, NJ 07806-5000."

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* 6.5 Delete the entire paragraph and substitute the following:

"6.5 Submission of ballistic test data and ammunition data cards. In addition to the normal distribution of records for items procured by the Department of the Army, two copies of all ballistic test data and ammunition data cards shall be forwarded to Commanding Officer, TACOM-ARDEC, ATTN: AMSTA-AR-QAA-R, Picatinny Arsenal, NJ 07806-5000."

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* 6.12 Change paragraph number to "6.14."

* 6.13 Change paragraph number to "6.15."

* Add new paragraph 6.12

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"6.12 Drawings. Drawings listed in section 2 of this specification under the heading U.S. Army Armament Research Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankfort Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC."

* Add new paragraph 6.13

"6.13 Definitions.

Acceptance Inspection Equipment (AIE). Any measuring device that is traceable to the national or international standard used to assure conformance of material to the contract requirements."

The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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